

Pattern recognition

COMPLETE LEARNING GUIDE

What You'll Learn

Pattern recognition questions — figure analogies, classification (odd-figure-out), matrix completion, figure counting, embedded figures, and rule-based rotation/reflection sequences — test visual-logical reasoning independent of language or numbers, evaluating whether a candidate can identify the single consistent transformation or shared attribute governing a set of figures. As with number series, the reliable approach is to explicitly name the rule (rotation by X degrees, addition of a line, a shared count of sides) rather than relying on a figure simply 'looking right,' since distractor options are deliberately designed to look plausible at a glance while violating the actual rule. This sub-module works through each major pattern-recognition question type with a concrete, fully reasoned example, including two solvable figure-counting problems worked with an exact count.

1. Figure Analogies

Figure analogies (Figure A : Figure B :: Figure C : ?) require identifying the exact transformation from A to B (a rotation, an added element, a color/shading change, a size change) and applying that same transformation to C — exactly like word analogies, but with a visual transformation instead of a semantic relationship.

2. Classification (Odd Figure Out)

Classification questions give several figures where all but one share a specific attribute (number of sides, number of lines of symmetry, presence of curved vs straight lines) — identify the shared attribute among the majority first, then find which figure breaks it.

- Four figures have 4, 4, 4, and 3 sides respectively. The shared attribute among three of them is '4 sides'; the figure with 3 sides is the odd one out.

3. Matrix (3×3 Grid) Figure Completion

Matrix questions arrange figures in a 3×3 grid where a consistent rule operates across each row and/or column (e.g., rotation increasing by a fixed angle moving left to right, or an element count increasing by one moving top to bottom) — verify the rule against at least two complete rows or columns before predicting the missing figure, since a rule that fits only one row may be coincidental.

4. Figure Counting (Triangles, Lines)

Counting problems (how many triangles/lines/squares in a figure) require systematically counting the smallest individual units first, then all valid larger combinations formed by joining adjacent smaller units — undercounting combined regions is the most common error.

- Triangle ABC has a single line drawn from vertex A to a point D on side BC. This creates: triangle ABD, triangle ACD, and the original triangle ABC itself — 3 triangles in total, not just the 2 smaller ones.

5. Embedded Figures

Embedded figure questions ask you to find a simple shape (like a specific triangle or quadrilateral) hidden within the outline of a more complex figure — the reliable technique is to trace the simple shape's exact outline mentally and check if all of its edges exist somewhere within the complex figure's lines, ignoring extra lines that aren't part of that specific shape.

6. Rule-Based Rotation and Reflection Sequences

A sequence of figures rotating or reflecting by a fixed, consistent amount at each step is solved by measuring the exact angle or transformation between the first two figures, then applying that same fixed increment forward to find the next figure in the sequence.

- A shape rotates 45° clockwise at each step in a sequence. If the sequence shows positions at 0° , 45° , 90° , 135° , the next figure in the sequence must be positioned at 180° .

7. Grouping Figures by Shared Attribute

Beyond simple side-counting, classification questions can group by lines of symmetry, whether a figure is open or closed, or whether it's made of straight lines only vs includes a curve — always check MULTIPLE possible shared attributes before settling on one, since the first attribute you notice isn't always the one that correctly isolates the odd figure.

Common Pitfalls

- Picking a figure that 'looks similar' at a glance instead of explicitly naming and verifying the exact transformation rule.
- In matrix (3×3) questions, confirming a rule against only one row or column instead of checking it holds across at least two.
- Undercounting figure-counting problems by missing larger triangles/shapes formed by combining smaller adjacent ones.
- In classification questions, fixating on the first shared attribute noticed (like color) when the actual distinguishing rule is a different attribute (like symmetry or side count).
- In embedded-figure questions, allowing extra lines in the complex figure to distract from tracing the exact target shape's outline.

- Assuming a rotation/reflection sequence's increment from only the first two figures without confirming it holds for a third given figure too.

Key Takeaways

- Always name the exact transformation rule (rotation angle, added element, attribute change) rather than judging by visual similarity alone.
- Verify a matrix (3×3 grid) rule against at least two rows or columns before predicting the missing figure.
- In figure counting, count the smallest units first, then systematically count all larger combinations formed by adjacent units.
- Check multiple candidate shared attributes (sides, symmetry, open/closed, curved/straight) in classification questions before picking the odd one out.
- For embedded figures, trace the target shape's exact outline and verify every edge exists within the complex figure.
- Confirm a rotation/reflection sequence's fixed increment against more than just the first two figures when a third is given.

Learn More

IndiaBix — Non-Verbal Reasoning: Classification

<https://www.indiabix.com/non-verbal-reasoning/classification/>

GeeksforGeeks — Non-Verbal Reasoning Questions and Answers

<https://www.geeksforgeeks.org/non-verbal-reasoning-questions-and-answers/>